A certified lower bound $\mu \geq 0.380020$ for Erdős' minimum overlap problem

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Abstract

Let μ denote the limiting constant in Erdős' minimum overlap problem. We prove $\mu \geq 0.380020$, improving the previous certified lower bound 0.37912; the best known upper bound is 0.380868, so this closes about 51% of the remaining gap.

The improvement comes from a constraint family absent from the two preceding Fourier programmes. Writing $f + g = \mathbf{1}_{[-1,1]}$ and $\psi = 2f - 1$, taking autocorrelations gives the exact identity $M(x)+M(-x) = \frac{1}{2}\big(\Lambda(x) - C_\psi(x)\big)$ with $\Lambda = (2-|x|)_+$. Since C_ψ is a compactly supported autocorrelation, $\widehat{C_\psi}(\xi) = |\widehat{\psi}(\xi)|^2 \geq 0$ for *every real* ξ, whereas White's Fourier programme and its recent strengthening use only discrete Fourier-series frequencies. Their lattice constraints are a sparse subsample of a continuum family, and it is refinement of the frequency spacing below that lattice spacing, not extension of the frequency range, that moves the bound. The Boas–Kac–Krein factorisation theorem, sharpened by Ehm–Gneiting–Richards, explains why this family is strong: in dimension one an even compactly supported positive-definite function is exactly the autocorrelation of a *real* function of half support. The description of the even part of the overlap function is therefore lossless apart from the pointwise bound $|\psi| \leq 1$, which enters only through $|C_\psi| \leq \Lambda$.

The bound is certified by an explicit rational multiplier vector. The certificate carries no feasibility constraints beyond sign conditions, so verification reduces to exact rational arithmetic plus an interval upper bound for one integral of an explicit Lipschitz function; a self-contained verification script is provided.

1 Introduction

For $n \geq 1$ let $A \cup B = \{1, \ldots, 2n\}$ be a partition into two sets of size n, and let

$$M(n) \;=\; \min_{A,B} \, \max_{k\in\mathbb{Z}} \big|A \cap (B+k)\big|.$$

Erdős asked in 1955 [3] for the asymptotics of $M(n)$. Haugland [4] proved that $\mu := \lim_{n\to\infty} M(n)/n$ exists. Erdős' own bound was $\mu \geq 1/4$; Moser [7] obtained $\mu \geq \sqrt{4-\sqrt{15}} \approx 0.35639$. Moser and Murdeshwar [8] subsequently introduced the continuous overlap formulation used below. White [9] built a convex programme over its Fourier data and obtained $\mu \geq 0.379005$, which was recently improved to $\mu \geq 0.37912$ [10] by adjoining two positive-semidefinite Toeplitz constraints to White's programme. On the other side, step-function constructions give $\mu \leq 0.380868$ [14], after a sequence 0.380926 [5], 0.380924 [11], 0.380876 [12], 0.380871 [13].

Theorem 1.1. $\mu \geq 0.380020$.

The proof is a finite computation whose output is an explicit rational certificate; the certificate and an independent verifier are described in Section 7.

1.1 What is new

The earlier lower bounds of Erdős and Moser–Murdeshwar are not Fourier programmes. The relevant comparison for our new constraint is instead with the Fourier/convex programmes of White [9] and Kim–Pilanci [10]. White expands the overlap function in the period-4 sine–cosine basis on $[-2,2]$. Its coefficient index $k \in \mathbb{Z}$ represents, in our full-line normalisation, the frequency $\xi = k/4$. Kim–Pilanci take White's programme as their base and add $T_f \succeq 0$ and $I - T_f \succeq 0$, Toeplitz constraints built from a discrete sequence of trigonometric moments of f. Their observation that adding more Fourier modes "refines the discretization but does not, in itself, tighten the relaxation" concerns enlargement along these fixed Fourier lattices.

Our novelty claim is correspondingly narrower: to our knowledge, neither of these Fourier-program lower bounds imposes necessary constraints at off-lattice frequencies $\xi \notin \frac{1}{4}\mathbb{Z}$.

Our observation is that the natural object is not periodic. Section 3 shows that $M(x) + M(-x)$ is, up to an explicit affine term, the autocorrelation of the single function $\psi = 2f - 1$ on $[-1,1]$. That autocorrelation is compactly supported, so its Fourier transform on the *whole line* is nonnegative, while the period-4 Fourier data sample only the lattice $\xi \in \frac{1}{4}\mathbb{Z}$. Numerically, refinement below this lattice spacing is what produces the gain, while extending the frequency range does essentially nothing.

Moreover, the Boas–Kac–Krein factorisation theorem [1] says that a continuous positive-definite function C supported in $[-2,2]$ has a representation

$$C(x) = \int u(t)\overline{u(t+x)}\,dt$$

for some generally complex u supported in $[-1,1]$. The root we need must be real, which is a genuinely separate question: Ehm, Gneiting and Richards [2, Thm. 2.1(b)] settle it in dimension one, showing that a real-valued Boas–Kac root exists if and only if C is even, as it is here. (Their results turn negative for radial functions in dimension $d \geq 2$, so the point needs checking rather than assuming.) Hence $\widehat{C} \geq 0$ together with the support condition is an exact description of the achievable C_ψ. What it does *not* impose is $|\psi| \leq 1$, whose even-part consequence used below is the two-sided bound $|C_\psi| \leq \Lambda$. We use the theorem only to explain the strength of the continuum family, not in the proof of Theorem 1.1.

1.2 Shape of the argument

Fix an admissible f and write $\Omega = \|M\|_\infty$ and $E = \int t f(t)\,dt$. We isolate seven exact facts about M (Section 4): three moment identities, two pointwise bounds, the continuum positive-definiteness family, and a family of linear consequences of a quadratic Fourier coupling between the even and odd parts of M. Each fact is linear in the pair $(M(x), M(-x))$ for fixed x. A nonnegative combination of them yields, for any choice of multipliers, an inequality of the form $V \leq \Sigma \leq 2\int_0^2 h$, where Σ is a single integral against M, V is an explicit constant, and h is the pointwise maximum of a linear function over an explicit pentagon (Section 6). If V exceeds $2\int_0^2 h$ then no admissible f with the given data exists.

The parameter E is eliminated by a branch and bound over a compact interval, using the bathtub principle to confine E (Section 5).

2 Notation and the reduction

Throughout, $f : [-1,1] \to [0,1]$ is measurable with $\int_{-1}^1 f = 1$, and $g = (1-f)\mathbf{1}_{[-1,1]}$, so that $g : \mathbb{R} \to [0,1]$, $\int g = 1$ and $f + g = \mathbf{1}_{[-1,1]}$ on $[-1,1]$. Set

$$M(x) \;=\; \int_{\mathbb{R}} f(t)\,g(x+t)\,dt, \qquad x \in \mathbb{R},$$

supported in $[-2,2]$, and $\Omega = \|M\|_\infty$. The following lemma is the only link we need between μ and this continuous class. The continuous formulation goes back to Moser–Murdeshwar [8]; we include the short proof because the displayed direction is the only one used in this paper.

Lemma 2.1. *$\mu \geq \inf_f \|M\|_\infty$, the infimum being over the admissible class above.*

Proof. Fix n and a partition $A \cup B = \{1,\dots,2n\}$ with $|A| = |B| = n$. Put $\delta = 1/n$ and $I_i = \big[-1+(i-1)\delta,\ -1+i\delta\big)$ for $i = 1,\dots,2n$, so that the I_i tile $[-1,1)$. Let $f = \sum_{i\in A} \mathbf{1}_{I_i}$. Then f takes values in $\{0,1\} \subset [0,1]$ and $\int f = n\delta = 1$, so f is admissible, and $g = \sum_{j\in B} \mathbf{1}_{I_j}$. For two intervals of equal length δ one has $|I_i \cap (I_j - x)| = \big(\delta - |x-(j-i)\delta|\big)_+$, so

$$M(x) \;=\; \sum_{i\in A} \sum_{j\in B} \big(\delta - |x-(j-i)\delta|\big)_+.$$

This is continuous, compactly supported and piecewise linear with every breakpoint in $\delta\mathbb{Z}$, hence attains its maximum at a breakpoint. At $x = k\delta$ every summand vanishes except those with $j - i = k$, each contributing δ; and $i \in A$ with $i + k \in B$ says exactly $i \in A \cap (B-k)$. Therefore

$$\|M\|_\infty \;=\; \max_{k\in\mathbb{Z}} M(k\delta) \;=\; \max_{k\in\mathbb{Z}} \frac{\big|A\cap(B-k)\big|}{n} \;=\; \frac{1}{n}\max_{k\in\mathbb{Z}}\big|A\cap(B+k)\big|.$$

Choosing A, B optimal for $M(n)$ gives $\inf_f \|M\|_\infty \leq M(n)/n$ for every n; let $n \to \infty$ and use Haugland's theorem [4] that the limit exists. □

Only this inequality is used below. We work with the continuous problem throughout. Write

$$\overline{M}(x) = \tfrac{1}{2}\big(M(x)+M(-x)\big), \qquad O(x) = \tfrac{1}{2}\big(M(x)-M(-x)\big),$$

so $\overline{M}$ is even, O is odd, and $M = \overline{M} + O$. Put

$$\psi = (2f-1)\mathbf{1}_{[-1,1]}, \qquad C_\psi(x) = \int_{\mathbb{R}} \psi(t)\,\psi(t+x)\,dt, \qquad \Lambda(x) = (2-|x|)_+.$$

We use $\widehat{h}(\xi) = \int_{\mathbb{R}} h(x)e^{-2\pi i\xi x}\,dx$ and set

$$S(\xi) \;=\; \widehat{\mathbf{1}_{[-1,1]}}(\xi) \;=\; \frac{\sin(2\pi\xi)}{\pi\xi}, \qquad S(0) = 2,$$

so that $\widehat{\Lambda} = S^2$. Finally $E = \int_{-1}^{1} t f(t)\,dt$.

Since $\overline{M}$ is even and real, $\widehat{\overline{M}}(\xi) = \int \overline{M}(x)\cos(2\pi\xi x)\,dx$ is real and even; since O is odd and real we write $\widetilde{O}(\xi) = \int_{\mathbb{R}} O(x)\sin(2\pi\xi x)\,dx$, so $\widehat{M} = \widehat{\overline{M}} - i\widetilde{O}$.

3 The autocorrelation identity

Lemma 3.1. *For all $x \in \mathbb{R}$,*

$$M(x) + M(-x) \;=\; \tfrac{1}{2}\big(\Lambda(x) - C_\psi(x)\big), \qquad \textit{equivalently} \qquad 4\overline{M} \;=\; \Lambda - C_\psi.$$

Proof. For $u, v \in L^2(\mathbb{R})$ write $(u \star v)(x) = \int u(t)v(x+t)\,dt$. Then $M(x) = (f\star g)(x)$ and $M(-x) = (g \star f)(x)$, and $\star$ is bilinear. Applying it to $f + g = \mathbf{1}_{[-1,1]}$ gives

$$\Lambda = \mathbf{1}_{[-1,1]} \star \mathbf{1}_{[-1,1]} = (f\star f) + (g\star g) + (f\star g) + (g\star f) = C_f + C_g + M(x) + M(-x),$$

while $\psi = f - g$ gives

$$C_\psi = (f-g)\star(f-g) = C_f + C_g - (f\star g) - (g\star f) = C_f + C_g - M(x) - M(-x).$$

Subtracting the second display from the first yields $\Lambda - C_\psi = 2\big(M(x)+M(-x)\big) = 4\overline{M}(x)$. □

The identity reduces the problem to a single function. Since $\int \psi = 0$ forces $\int C_\psi = \widehat{C_\psi}(0) = 0$, one has $\int(\Lambda - C_\psi) = \int \Lambda = 4$, recovering $\int M = 1$; and the whole difficulty of the problem is concentrated in the single constraint $\widehat{\psi}(0) = 0$, since $\psi \equiv 1$ would make $\Lambda - C_\psi$ vanish identically.

Remark 3.2. Lemma 3.1 also makes White's key structural property immediate. Since $\widehat{\Lambda}(\xi) = S(\xi)^2$ vanishes at every nonzero half-integer, evaluating $4\widehat{\overline{M}} = S^2 - |\widehat{\psi}|^2$ at $\xi = m/2$ gives $\widehat{\overline{M}}(m/2) = -\frac{1}{4}|\widehat{\psi}(m/2)|^2 \le 0$ for every nonzero integer m, which is White's condition $A_{2m} \le 0$; likewise $\widetilde{O}(m/2) = 0$ is his $B_{2m} = 0$.

4 Exact constraints on the overlap function

Lemma 4.1. $\displaystyle\int_{\mathbb{R}} \overline{M} = 1$, $\displaystyle\int_{\mathbb{R}} x^2\overline{M}(x)\,dx = \tfrac{2}{3} + 2E^2$, *and* $\displaystyle\int_{\mathbb{R}} x\,O(x)\,dx = -2E$.

Proof. Both f and g are probability densities, and M is the density of $U - T$ for independent $T \sim f$, $U \sim g$: indeed $\int f(t)g(x+t)\,dt$ is exactly that density. Hence $\int M = 1$. Since $f + g = \mathbf{1}_{[-1,1]}$ we get $\int t\,g(t)\,dt = -E$ and $\int t^2 f + \int t^2 g = \int_{-1}^{1} t^2\,dt = \frac{2}{3}$. Therefore

$$\int xM(x)\,dx = \mathbb{E}[U] - \mathbb{E}[T] = -2E, \qquad \int x^2 M(x)\,dx = \mathbb{E}[U^2] + \mathbb{E}[T^2] - 2\,\mathbb{E}[U]\mathbb{E}[T] = \tfrac{2}{3} + 2E^2.$$

Splitting into even and odd parts gives the three claims, since $\int \overline{M} = \int M$, $\int x^2\overline{M} = \int x^2 M$ and $\int x\,O = \int x\,M$. □

Lemma 4.2. *For every x, $0 \le M(x) \le \Omega$ and*

$$M(x) + M(-x) \;\le\; 2 - |x| \qquad (|x| \le 2).$$

Proof. The first is the definition of Ω together with $f, g \ge 0$. For the second, let $I_x = [-1,1] \cap ([-1,1] - x)$, of length $2 - |x|$. Then

$$M(x) + M(-x) = \int_{I_x} \Big(f(t)\big(1 - f(t+x)\big) + f(t+x)\big(1 - f(t)\big)\Big)dt = \int_{I_x} \big(a + b - 2ab\big)dt$$

with $a = f(t)$, $b = f(t+x)$ in $[0,1]$. Since $1 - a - b + 2ab = (1-a)(1-b) + ab \ge 0$, the integrand is at most 1, whence the bound. □

Note that this is strictly stronger than bounding each of $M(\pm x)$ by $2 - |x|$ separately.

Lemma 4.3 (continuum positive definiteness)**.** *For every real ξ,*

$$\int_{\mathbb{R}} \overline{M}(x)\cos(2\pi\xi x)\,dx \;\le\; \tfrac{1}{4}S(\xi)^2.$$

Proof. By Lemma 3.1, $4\widehat{\overline{M}} = \widehat{\Lambda} - \widehat{C_\psi} = S^2 - |\widehat{\psi}|^2 \le S^2$, and $\widehat{\overline{M}}(\xi)$ is the stated cosine integral. □

Lemma 4.4 (even/odd coupling)**.** *For every real ξ and every real τ,*

$$4S(\xi)^2\,\widehat{\overline{M}}(\xi) \;+\; 8\tau\,\widetilde{O}(\xi) \;-\; 4\tau^2 \;\le\; S(\xi)^4.$$

Proof. Since f is real, $\widehat{M}(\xi) = \overline{\widehat{f}(\xi)}\,\widehat{g}(\xi)$: substituting $y = x+t$, $\iint f(t)g(x+t)e^{-2\pi i\xi x}\,dt\,dx = \left(\int f(t)e^{2\pi i\xi t}dt\right)\left(\int g(y)e^{-2\pi i\xi y}dy\right)$. From $f = \frac{1}{2}(\mathbf{1}_{[-1,1]} + \psi)$ and $g = \frac{1}{2}(\mathbf{1}_{[-1,1]} - \psi)$ we get $\widehat{f} = \frac{1}{2}(S + \widehat{\psi})$ and $\widehat{g} = \frac{1}{2}(S - \widehat{\psi})$, so with $\widehat{\psi} = P + iQ$ (P, Q real),

$$\widehat{M} = \tfrac{1}{4}\big(S + \overline{\widehat{\psi}}\big)\big(S - \widehat{\psi}\big) = \tfrac{1}{4}\Big(S^2 - P^2 - Q^2 - 2iSQ\Big).$$

Comparing with $\widehat{M} = \widehat{\widehat{M}} - i\widetilde{O}$ gives $4\widehat{\widehat{M}} = S^2 - P^2 - Q^2$ and $\widetilde{O} = \frac{1}{2}SQ$. Hence

$$4S^2\widehat{\widehat{M}} = S^4 - S^2P^2 - S^2Q^2 = S^4 - S^2P^2 - 4\widetilde{O}^2 \;\leq\; S^4 - 4\widetilde{O}^2.$$

Finally $4\widetilde{O}^2 \geq 8\tau\widetilde{O} - 4\tau^2$, because $(2\widetilde{O} - 2\tau)^2 \geq 0$. □

Lemma 4.4 is the only place where a quadratic constraint enters, and the tangent-line form makes it linear at the cost of a free parameter τ; *any* real τ gives a valid inequality, so no feasibility question arises when a numerical value of τ is supplied.

Remark 4.5. Lemmas 4.3 and 4.4 are not independent. Taking $\tau = 0$ in Lemma 4.4 gives $4S(\xi)^2\widehat{\widehat{M}}(\xi) \leq S(\xi)^4$, which for $S(\xi) \neq 0$ is exactly Lemma 4.3; so off the zero set of S the positive-definiteness family is the $\tau = 0$ member of the coupling family, and Lemma 4.4 is the stronger statement. Lemma 4.3 carries independent content precisely at the nonzero half-integers, where S vanishes and Lemma 4.4 degenerates: there it forces $\widetilde{O}(\xi) = 0$ (divide by τ and let $\tau \to 0^{\pm}$) but says nothing about $\widehat{\widehat{M}}$, whereas Lemma 4.3 still gives $\widehat{\widehat{M}}(\xi) \leq 0$. We keep both as separate facts because the resulting linear programme is better conditioned, and because the two families are used very differently by the certificates of Section 7: on the binding box $E \in [0, 0.0269]$, which is the box that determines Theorem 1.1, the multipliers γ_j are numerically negligible — together they contribute under 10^{-7} to G_e and to G_o, against $\sum_j c_j \approx 1.61$ — so the bound is carried entirely by the c_j, whereas on the remaining boxes γ contributes substantially. This is what one expects, since $\widetilde{O}$ is small when E is small and Lemma 4.4 constrains only $\widetilde{O}$ beyond what Lemma 4.3 already gives.

5 Confining the first moment

Lemma 5.1 (bathtub bound)**.** *If ρ is a probability density on $\mathbb{R}$ with $\rho \leq \Omega$ a.e., then* $\mathrm{Var}(\rho) \geq \frac{1}{12\Omega^2}$.

Proof. Let m be the mean. Replacing ρ by its symmetric decreasing rearrangement about m does not increase $\int (x-m)^2\rho\,dx$ and preserves both the mass and the bound $\rho \leq \Omega$. Among symmetric decreasing densities bounded by Ω the second moment about m is minimised by $\Omega\mathbf{1}_{[m-1/(2\Omega),\,m+1/(2\Omega)]}$ (bathtub principle, [6, Thm. 1.14]), whose variance is $\frac{1}{12\Omega^2}$. □

Corollary 5.2. *Any admissible f satisfies $E^2 \leq \frac{1}{3} - \frac{1}{24\Omega^2}$.*

Proof. M is a probability density bounded by Ω, and by Lemma 4.1 $\mathrm{Var}(M) = \left(\frac{2}{3} + 2E^2\right) - (2E)^2 = \frac{2}{3} - 2E^2$. Apply Lemma 5.1. □

Replacing $f(t)$ by $f(-t)$ replaces $M(x)$ by $M(-x)$, leaving Ω unchanged and sending $E \mapsto -E$; so we may and do assume $E \geq 0$.

6 The certificate

Fix an interval $[E_-, E_+] \subseteq [0, \infty)$, a level $\Omega > 0$, finitely many frequencies $\xi_1, \dots, \xi_m > 0$ with tangent points $\tau_1, \dots, \tau_m \in \mathbb{R}$, and multipliers

$$\lambda \in \mathbb{R}, \qquad \nu_+,\ \nu_-,\ \kappa_+,\ \kappa_-,\ c_1, \dots, c_m,\ \gamma_1, \dots, \gamma_m\ \geq\ 0.$$

Put $\nu = \nu_+ - \nu_-$, $\kappa = \kappa_+ - \kappa_-$, $S_j = S(\xi_j)$, and define on $\mathbb{R}$

$$G_e(x) = \lambda + \nu x^2 - \sum_{j=1}^m c_j \cos(2\pi\xi_j x) - \sum_{j=1}^m 4\gamma_j S_j^2 \cos(2\pi\xi_j x),$$
$$G_o(x) = \kappa x - \sum_{j=1}^m 8\gamma_j \tau_j \sin(2\pi\xi_j x),$$

and

$$V\ =\ \lambda + \nu_+\Big(\tfrac{2}{3} + 2E_-^2\Big) - \nu_-\Big(\tfrac{2}{3} + 2E_+^2\Big) - \sum_j \tfrac{1}{4} c_j S_j^2 - 2\kappa_+ E_+ + 2\kappa_- E_- - \sum_j \gamma_j\big(S_j^4 + 4\tau_j^2\big).$$

Finally let $\alpha = \frac{1}{2}(G_e + G_o)$, $\beta = \frac{1}{2}(G_e - G_o)$ and, for $x \in (0, 2)$,

$$h(x)\ =\ \max\Big\{\ \alpha(x)p + \beta(x)q\ :\ 0 \leq p, q \leq \Omega,\ p + q \leq 2 - x\ \Big\}.$$

The feasible set is a polygon with at most five vertices, namely $(0, 0)$, $(A, 0)$, $(0, A)$ and, when $2 - x > \Omega$, (Ω, B) and (B, Ω), where $A = \min(\Omega, 2 - x)$ and $B = \min\big(\Omega, (2 - x - \Omega)_+\big)$; so h is the maximum of at most five explicit functions.

Theorem 6.1. *With the notation above, if*

$$V\ >\ 2\int_0^2 h(x)\,dx,$$

then there is no admissible f with $E \in [E_-, E_+]$ and $\|M\|_\infty \leq \Omega$.

Proof. Suppose such an f exists and set $\Sigma = \int_{-2}^2 \big(\overline{M} G_e + O\, G_o\big)$, which is finite.

Lower bound. Expanding G_e and G_o and using $\overline{M}$ even, O odd,

$$\Sigma = \lambda \int \overline{M} + \nu \int x^2 \overline{M} - \sum_j c_j \widehat{\overline{M}}(\xi_j) - \sum_j \gamma_j \Big(4S_j^2 \widehat{\overline{M}}(\xi_j) + 8\tau_j \widetilde{O}(\xi_j)\Big) + \kappa \int x\, O.$$

By Lemma 4.1 the first, second and last terms equal λ, $\nu(\frac{2}{3} + 2E^2)$ and $-2\kappa E$. Since $\nu_\pm \geq 0$ and $E_-^2 \leq E^2 \leq E_+^2$, $\nu(\frac{2}{3} + 2E^2) \geq \nu_+(\frac{2}{3} + 2E_-^2) - \nu_-(\frac{2}{3} + 2E_+^2)$; since $\kappa_\pm \geq 0$ and $E_- \leq E \leq E_+$, $-2\kappa E \geq -2\kappa_+ E_+ + 2\kappa_- E_-$. By Lemma 4.3 and $c_j \geq 0$, $-c_j \widehat{\overline{M}}(\xi_j) \geq -\frac{1}{4} c_j S_j^2$; by Lemma 4.4 and $\gamma_j \geq 0$, $-\gamma_j(4S_j^2 \widehat{\overline{M}} + 8\tau_j \widetilde{O}) \geq -\gamma_j(S_j^4 + 4\tau_j^2)$. Summing, $\Sigma \geq V$.

Upper bound. $\overline{M} G_e + O G_o$ is even, so $\Sigma = 2\int_0^2 (\overline{M} G_e + O G_o)$. For fixed $x \in (0, 2)$ put $p = M(x)$, $q = M(-x)$, so that $\overline{M} = \frac{1}{2}(p + q)$, $O = \frac{1}{2}(p - q)$ and

$$\overline{M}(x) G_e(x) + O(x) G_o(x)\ =\ \alpha(x)\, p + \beta(x)\, q.$$

By Lemma 4.2, (p, q) lies in the polygon defining $h(x)$, so the integrand is at most $h(x)$ pointwise and $\Sigma \leq 2\int_0^2 h$. Combining, $V \leq \Sigma \leq 2\int_0^2 h < V$, a contradiction. □

Two features of Theorem 6.1 matter for verification. First, the multipliers are subject to *no* constraint other than the stated signs: every choice yields a true implication, so a numerically computed multiplier vector needs only to be rounded to rationals and sign-checked. This is what replaces the dual-feasibility check that a semidefinite formulation would require. Second, the only nontrivial quantity is $\int_0^2 h$, an integral of an explicit piecewise-smooth function.

Lemma 6.2. *h is Lipschitz on $[0,2]$ with constant*

$$L \;=\; \Omega\Big(\sup|G_e'| + \sup|G_o'|\Big) + \sup|G_e| + \sup|G_o|,$$

and consequently, for the midpoint rule on N equal cells of width $\delta = 2/N$,

$$\int_0^2 h \;\le\; \delta\sum_{i=1}^{N} h\big(x_i\big) \;+\; \tfrac{1}{2}L\,\delta, \qquad x_i = (i-\tfrac{1}{2})\delta.$$

Proof. Each candidate function is $\phi_v = \alpha P_v + \beta Q_v$ with $|P_v|, |Q_v| \le \Omega$ and $|P_v'|, |Q_v'| \le 1$ (each of A, B, Ω, 0 is piecewise linear with slope in $\{0,-1\}$). Hence $|\phi_v'| \le \Omega(|\alpha'|+|\beta'|)+(|\alpha|+|\beta|) \le \Omega(|G_e'| + |G_o'|) + |G_e| + |G_o|$, using $|\alpha| + |\beta| \le |G_e| + |G_o|$ and likewise for the derivatives. A pointwise maximum of functions with a common Lipschitz constant has that constant, and the constant function 0 is included. The midpoint estimate for a Lipschitz function is $\big|\int_I h - |I|h(\operatorname{mid} I)\big| \le \frac{1}{4}L|I|^2$; summing over N cells gives $\frac{1}{4}LN\delta^2 = \frac{1}{2}L\delta$. □

Explicit suprema follow from $|x| \le 2$ and $|\cos|, |\sin| \le 1$:

$$\sup|G_e| \le |\lambda| + 4|\nu| + \textstyle\sum_j c_j + \sum_j 4\gamma_j S_j^2, \qquad \sup|G_e'| \le 4|\nu| + \sum_j 2\pi\xi_j c_j + \sum_j 8\pi\xi_j\gamma_j S_j^2,$$
$$\sup|G_o| \le 2|\kappa| + \textstyle\sum_j 8\gamma_j|\tau_j|, \qquad \sup|G_o'| \le |\kappa| + \sum_j 16\pi\xi_j\gamma_j|\tau_j|.$$

7 The computation

7.1 Choosing the multipliers

For a fixed box $[E_-, E_+]$ the optimal multipliers are obtained from a linear programme: normalise $V = 1$ and minimise $2\int_0^2 h$, discretising x on a uniform grid and introducing one variable per grid point to represent h. This is exactly the dual of the natural primal relaxation, and we use the agreement of the two as a consistency check. The comparison has to be set up with care: it is meaningful only at a single fixed E, since over a nondegenerate box the dual must hold for every E in the box and therefore falls strictly below the primal value at any one of them; and it must use the same frequency set and the same x-grid on both sides. Under those conditions, taking E at the midpoint of the binding box, the two agree to 6.4×10^{-7} at spacing $1/16$ and to 2.4×10^{-6} at the spacing $1/64$ used here. The tangent points τ_j are read off from a solution of that primal relaxation. *The grid used here affects only the quality of the multipliers, never the validity of the conclusion*, which is governed solely by Theorem 6.1.

We use $m = 768$ frequencies $\xi_j = j/64$, $j = 1, \ldots, 768$, i.e. spacing $1/64$ up to $\xi = 12$. Such a frequency lies on White's period-4 lattice $\frac{1}{4}\mathbb{Z}$ exactly when $16 \mid j$; fifteen sixteenths of the frequencies are off-lattice.

7.2 Branch and bound over E

By Corollary 5.2 and the symmetry $E \mapsto -E$, it suffices to treat $E \in [0, E_{\max}]$ with $E_{\max}^2 = \frac{1}{3} - \frac{1}{24\Omega^2}$. The verifier checks that the boxes tile $[0, E_{\max}]$: that they start at 0, leave no gap, and reach past $E_{\max}$. Table 1 lists the boxes and the level certified on each; the reported bound is the minimum.

7.3 Rigorous verification

Given a rational multiplier vector, the verification consists of:

1. checking the sign conditions exactly on the rationals;
2. bounding $S(\xi_j)^2 = \sin^2(2\pi\xi_j)/(\pi\xi_j)^2$ from above and below by rationals. For $\sin^2 t = \frac{1}{2}(1 - \cos 2t)$, so no sign handling is needed, and for $\xi_j = j/64$ the angle $4\pi\xi_j$ is an exact integer multiple of $2\pi/64$; the reduction to the first octant is therefore exact integer arithmetic, and the residual angle is enclosed by an alternating Taylor series with an explicit truncation bound, in rational arithmetic rounded outwards to a fixed denominator. The only irrational input is π, enclosed from Machin's formula with an explicit alternating-series truncation bound, so no numerical constant is taken on trust. As an independent check the verifier confirms that this general routine reproduces the closed-form values 0, $\frac{2-\sqrt{2}}{4}$, $\frac{1}{2}$, $\frac{2+\sqrt{2}}{4}$, 1 available at spacing 1/16, from a rational enclosure of $\sqrt{2}$ that it checks by squaring;
3. evaluating V in exact rational arithmetic, using the upper enclosures of S_j^2 wherever S_j^2 occurs with a nonpositive coefficient, so that the computed V is a rigorous lower bound for the true V;
4. bounding $\int_0^2 h$ via Lemma 6.2 on $N = 2\,000\,000$ cells. Since $\xi_j = j/64$, every trigonometric argument is $j\theta$ with $\theta = \pi x/32$. Exact integer range reduction maps each $j\theta$ to an angle in $[0, \pi/4]$. On that interval the verifier encloses sine and cosine between alternating Taylor partial sums, with every elementary binary64 operation rounded outwards using `nextafter`. It then evaluates G_e, G_o by interval arithmetic, takes an interval upper bound for each value of h, and accumulates the nonnegative grid sum by upward-rounded pairwise addition. Thus no library sine or cosine evaluation, BLAS reduction, or asserted floating-point slack enters the final inequality. A faster ordinary floating-point grid is used only to propose the six-decimal level that is subsequently checked by this interval path.

7.4 Results

E-box	certified level	certified margin
[0.00000, 0.02690]	0.380020	$+1.83 \times 10^{-5}$
[0.02690, 0.05379]	0.382780	$+5.60 \times 10^{-6}$
[0.05379, 0.08069]	0.392960	$+2.02 \times 10^{-5}$
[0.08069, 0.10758]	0.399990	$+9.75 \times 10^{-3}$
[0.10758, 0.16137]	0.399990	$+2.49 \times 10^{-2}$
[0.16137, 0.21516]	0.399990	$+7.35 \times 10^{-2}$

Table 1: Certified level on each box; the bound of Theorem 1.1 is the minimum. The margin is V minus the *rigorous interval upper bound* for $2\int_0^2 h$, so it already includes the Lipschitz discretisation term $L\delta$ of Lemma 6.2.

Proof of Theorem 1.1. Write $\Omega^* = 0.380020$ and suppose some admissible f had $\Omega = \|M\|_\infty \le \Omega^*$. By the symmetry recorded after Corollary 5.2 we may assume $E \ge 0$. Since $\int M = 1$ and M is supported in $[-2, 2]$ we have $\Omega \ge \frac{1}{4} > 0$, so Corollary 5.2 applies and gives $E^2 \le \frac{1}{3} - \frac{1}{24\Omega^2}$. The map $t \mapsto \frac{1}{3} - \frac{1}{24t^2}$ is increasing on $(0, \infty)$, so $\Omega \le \Omega^*$ yields

$$E^2 \le \tfrac{1}{3} - \frac{1}{24(\Omega^*)^2} < 0.04481339, \qquad \text{that is} \qquad 0 \le E \le 0.211692.$$

The six boxes of Table 1 begin at 0, are consecutive, and end at 0.215164, so they tile $[0, 0.215164] \supset [0, 0.211692]$ and E lies in one of them, say the i-th, with certified level Ω_i. For that box the hypothesis $V > 2\int_0^2 h$ of Theorem 6.1 is checked at level Ω_i by the verification of Section 7, so no admissible f has E in that box together with $\|M\|_\infty \le \Omega_i$. The hypothesis of Theorem 6.1 is the inequality $\|M\|_\infty \le \Omega_i$ rather than an equality, so a certificate at level Ω_i excludes every smaller level as well; and $\Omega \le \Omega^* = \min_i \Omega_i \le \Omega_i$. This contradicts the existence of f.

Hence every admissible f satisfies $\|M\|_\infty > \Omega^*$, so $\inf_f \|M\|_\infty \ge \Omega^*$, and $\mu \ge \Omega^* = 0.380020$ by Lemma 2.1. □

8 Discussion

8.1 Where the remaining loss is

By Boas–Kac–Krein [1], a continuous positive-definite function supported in $[-2, 2]$ is a Hermitian autocorrelation with a generally complex root supported in $[-1, 1]$; and by Ehm–Gneiting–Richards [2, Thm. 2.1(b)], in dimension one that root may be taken *real* precisely when the function is even, as C_ψ is. The continuum family of Lemma 4.3, imposed at all real ξ together with the two moment identities, is therefore an exact description of the achievable C_ψ, equivalently of the even part $\overline{M}$, up to the single remaining requirement $|\psi| \le 1$. That requirement implies the two-sided pointwise bound

$$|C_\psi(x)| \le \Lambda(x).$$

In the certificate, $C_\psi \le \Lambda$ is equivalent to $M(x) + M(-x) \ge 0$, while $C_\psi \ge -\Lambda$ is equivalent to the sum bound $M(x) + M(-x) \le \Lambda(x)$ of Lemma 4.2; both sides are used.

That statement concerns $\overline{M}$ alone, and it is worth being precise about what the programme as a whole still discards. There are four distinct sources of loss.

1. *The pointwise bound.* The requirement $|\psi| \le 1$ is used, for the even part, only through $|C_\psi| \le \Lambda$. A bounded root need not exist merely because some real L^2 root does. For the even part this is the only loss.

2. *Joint realisability across frequencies.* At a *single* ξ with $S(\xi) \ne 0$, Lemma 4.4 is sharp: as (P, Q) ranges over $\mathbb{R}^2$, the pair $\big(\widehat{\overline{M}}, \widetilde{O}\big) = \big(\frac{1}{4}(S^2 - P^2 - Q^2),\ \frac{1}{2}SQ\big)$ sweeps out exactly $\{4S^2\widehat{\overline{M}} + 4\widetilde{O}^2 \le S^4\}$. But the constraints at different ξ must be met by one common real ψ, and that coupling is nowhere imposed. Boas–Kac–Krein, even in its real form, constrains only the modulus data $|\widehat{\psi}|$; it does not couple the phase that, through $Q = \operatorname{Im}\widehat{\psi}$, carries $\widetilde{O}$.

3. *Linearisation.* The tangent-line step in Lemma 4.4 is exact only when $\tau_j = \widetilde{O}(\xi_j)$ at the extremiser; any other choice loses.

4. *Finitely many frequencies.* Only m members of a continuum family are imposed. The loss is real but visibly converging. On the binding box the linear programme returns 0.379863, 0.379996 and 0.380031 at spacings 1/16, 1/32 and 1/64: the first refinement gains 1.3×10^{-4}, the second only 3.5×10^{-5}. Theorem 1.1 is certified at spacing 1/64, and a further refinement of the grid looks unlikely to repay its cost.

The bound of Theorem 1.1 is therefore not the limit of the method, and refining the frequency grid — not only a better treatment of $|\psi| \le 1$ — still moves it. What Boas–Kac–Krein and [2] rule out is any gain from a *different* family of constraints on $\overline{M}$: support, positive definiteness and the two moments already capture the *real* half-support autocorrelation cone exactly, so a sharper bound on the even part would have to exploit boundedness of the root.

8.2 Towards the exact constant

Numerically the extremal f is far from a 0/1 function: at a near-optimal f with $\|M\|_\infty \approx 0.3818$ one finds $\int \psi^2 \approx 0.967$, against the value 2 that a 0/1 function would give, and about 70% of f takes values strictly inside $(0, 1)$. The maximum of M is *not* attained at the origin ($M(0) \approx 0.258$), and the contact set $A = \{x : M(x) = \|M\|_\infty\}$ appears to have positive measure. For $x \in A$, Lemma 3.1 gives only

$$C_\psi(x) \;=\; (2-|x|) - 2\Omega - 2M(-x), \qquad (2-|x|) - 4\Omega \;\le\; C_\psi(x) \;\le\; (2-|x|) - 2\Omega.$$

Thus one-sided contact does not by itself force C_ψ to be affine: the reflected value $M(-x)$ may vary. The stronger identity

$$C_\psi(x) \;=\; (2-|x|) - 4\Omega$$

holds precisely on the *two-sided contact set* $A \cap (-A)$.

For the numerical candidate, continuity and $M(0) < \Omega$ make A avoid a neighbourhood of 0. Consequently, if $A \cap (-A)$ contains an interval $[a, b] \subset (0, 2)$, then C_ψ agrees with $2 - x - 4\Omega$ there and, by evenness, with $2 - |x| - 4\Omega$ on the reflected interval. This is the rigidity statement suggested by the computation; positive measure of A alone is not enough.

Question 8.1. Can one prove that an extremiser has an interval $[a, b] \subset A \cap (-A)$ with $0 < a < b < 2$? If so, do the resulting affine identities for C_ψ on $[a, b]$ and $[-b, -a]$, together with $\int \psi = 0$ and $|\psi| \le 1$, determine ψ? An affirmative answer, together with existence, would help characterise the extremiser.

8.3 AI disclosure

Artificial intelligence was used extensively to generate this paper. The Paratelligent Research Agent was used to generate the proof as well as an initial draft of the paper. Paratelligent is an AI system designed and built by this paper's human author. It's an overall system where multiple AI agents work simultaneously on hard research problems. Additionally, Claude and Codex were both used directly to get the paper ready to share publicly.

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A Reproducibility

The certificate data and a self-contained verification script are available at

https://github.com/pickhardt/erdos-minimum-overlap-certificate

The script depends only on the Python standard library and `numpy`, derives π and $\sqrt{2}$ internally with proved enclosures, and prints the verified bound. At its default grid $N = 2 \times 10^6$ the interval verification takes about ten minutes on a laptop; a smaller N passed on the command line reaches the same conclusion more quickly, with a proportionally smaller margin. Its output on the accompanying certificate is reproduced below, with two numeric columns dropped to fit the page:

```
pi enclosed to width 7.03e-86 by Machin's formula (checked internally)
sqrt(2) enclosure checked by squaring
S^2 Taylor enclosures agree with closed-form sqrt(2) values (den=16 self-test)
search grid N = 2000000, final verification grid N = 2000000

  E in [0.00000,0.02690]  certified Omega = 0.380020  margin +1.83e-05  OK  (L=12.6)
  E in [0.02690,0.05379]  certified Omega = 0.382780  margin +5.60e-06  OK  (L=21.8)
  E in [0.05379,0.08069]  certified Omega = 0.392960  margin +2.02e-05  OK  (L=23.5)
  E in [0.08069,0.10758]  certified Omega = 0.399990  margin +9.75e-03  OK  (L=23.3)
  E in [0.10758,0.16137]  certified Omega = 0.399990  margin +2.49e-02  OK  (L=24.1)
  E in [0.16137,0.21516]  certified Omega = 0.399990  margin +7.35e-02  OK  (L=22.8)

  all boxes excluded: True
  E-range: any f with ||M||_inf <= 0.380020 has E^2 <= 0.04481338,
           i.e. |E| <= 0.211692; boxes tile [0,0.215164] with no gap: True   OK

  VERIFIED:  mu >= 0.380020
```